Day 2: Mastering the Linux Terminal & Essential Commands

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1. Introduction

In ethical hacking, the **Terminal (CLI)** is your most powerful weapon. While Windows and macOS rely heavily on a Graphical User Interface (GUI), a hacker performs 95% of their work using text-based commands. Today was all about moving away from the mouse and learning to talk directly to the operating system.

2. Objective of the Day

The goal of Day 2 is to become comfortable navigating the Linux file system, managing files and folders, and understanding the core commands required to run security tools in Kali Linux.

3. Why Linux for Hacking?

Linux is the preferred OS for cybersecurity because:

- **Open Source:** You can modify any part of the system.
- **Lightweight:** It runs efficiently on virtual machines.
- **Control:** Most hacking tools are built specifically for the Linux terminal.

4. Essential Command Categories

A. Navigation Commands

- `pwd` (Print Working Directory): Tells you exactly which folder you are in.
- `ls`: Lists all files and folders in the current directory.
    - `ls -la`: Shows hidden files and detailed info (permissions, size).
- `cd [folder]`: Change directory to move between folders.
    - `cd ..`: Moves you one step back (to the parent folder).

B. File & Directory Management

- `mkdir [name]`: Creates a new folder.
- `touch [name].txt`: Creates an empty text file.
- `cp [source] [destination]`: Copies a file.
- `mv [source] [destination]`: Moves or renames a file.
- `rm [file]`: Deletes a file (**Be careful! There is no "Recycle Bin" in the terminal**).
- `cat [file]`: Displays the content of a file on the screen.

C. System & Network Info

- `whoami`: Tells you which user account you are currently using.
- `top`: Shows running processes and RAM usage (like Task Manager).
- `clear`: Cleans up your terminal screen.

5. User Permissions & Sudo

Linux is built on permissions.

- **Standard User:** Has limited access.
- **Root User:** The "Superuser" who can do anything.
- **Sudo (SuperUser Do):** We use this prefix before a command to run it with administrative privileges (e.g., `sudo apt update`).

6. What I Did Today

1. **Terminal Immersion:** I spent the day exclusively using the terminal in Kali Linux instead of the file explorer.
2. **Folder Architecture:** I created a structured workspace for my internship tasks using `mkdir`.
3. **File Manipulation:** I practiced creating, reading, and moving logs and notes using `touch`, `cat`, and `mv`.
4. **Searching:** I used the `grep` command to search for specific text within a file, a vital skill for analyzing large log files later on.

7. What I Learned

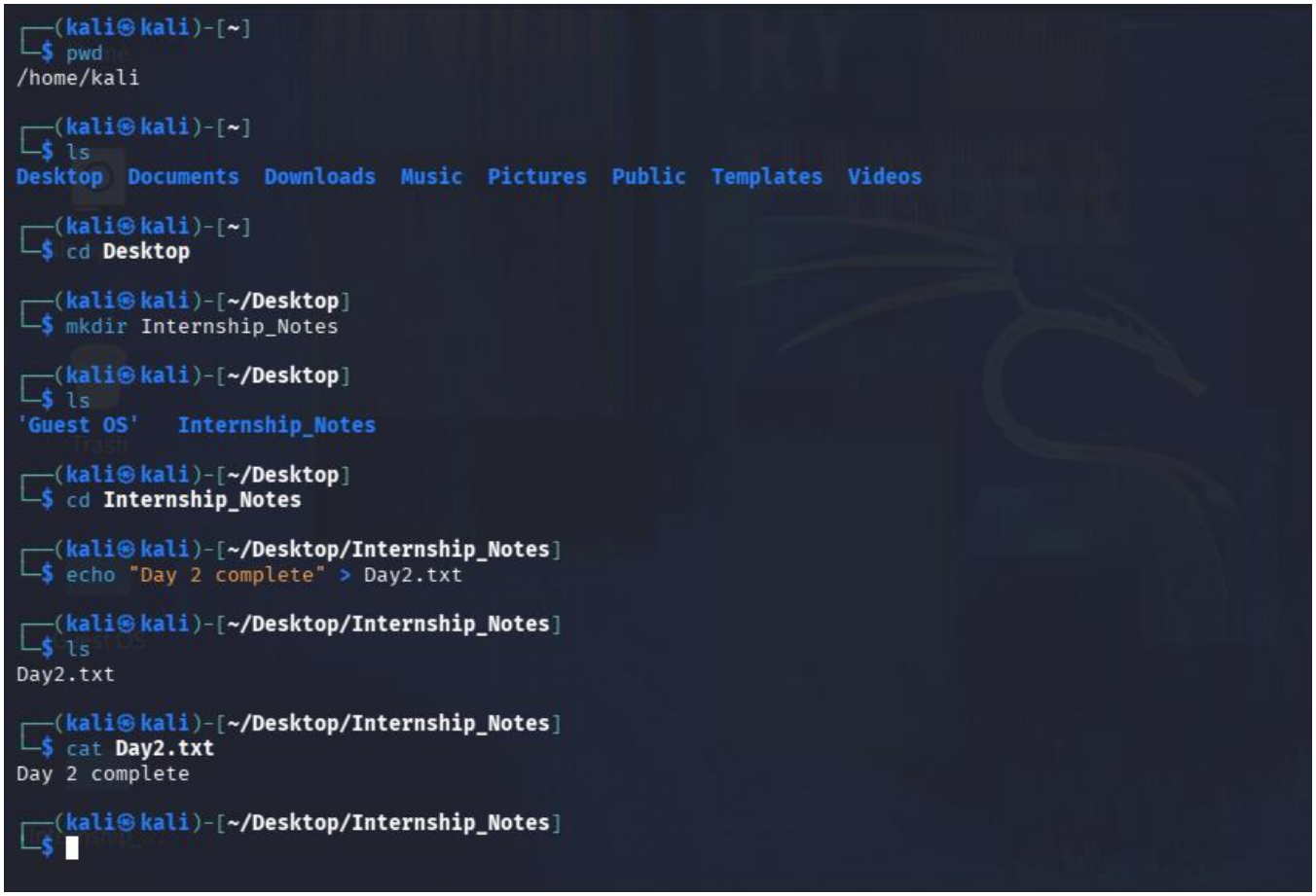
- **Speed & Efficiency:** I realized that typing a command is often much faster than clicking through five different folders.
- **Case Sensitivity:** Linux is case-sensitive. `Folder1` and `folder1` are two completely different things!
- **Tab Completion:** I learned the "Tab" key trick. Typing the first few letters of a command and hitting Tab completes it automatically—this saves time and prevents typos.
- **The Power of Man Pages:** I learned that typing `man [command]` (e.g., `man nmap`) opens a full manual for any tool.

Key Takeaway for Newbies:

"Don't be afraid of the black screen. The terminal isn't just a place to type codes; it's a direct conversation with the computer's brain."

8. Hands-on Examples & Practice

- **Task:** Create a folder named "Internship_Notes", go inside it, and create a file named "Day2.txt".

Execution:

1. `mkdir Internship_Notes`
2. `cd Internship_Notes`
3. `touch Day2.txt`
4. `ls` (to verify the file exists)